
Validating Causal Steering in Protein Language Models: Proxy Accuracy Does Not Guarantee Steering Success

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Abstract

1 Activation steering changes model output by modifying internal activations. We
2 ask whether agreement between a scoring proxy and experimental labels also
3 indicates that the proxy will provide a useful steering target. We test five protein
4 properties in ESM2-650M with difference-of-means steering vectors and a fixed
5 six-requirement evaluation protocol. The binding-affinity proxy has the strongest
6 held-out correlation in the study ($r = 0.80$), but its steering direction has no
7 measurable effect. A weaker catalytic-activity proxy ($r = 0.22$) produces an effect
8 that repeats across three random seeds. We find a similar mismatch for individual
9 attention heads: strong correlation with known 3D contacts does not imply a large
10 effect under ablation. Two additional checks change how we interpret other results.
11 A residue-exclusion check gives different conclusions across seeds, and an apparent
12 expression-yield effect uses a direction that overlaps with the intrinsic-disorder
13 direction. These results show that proxy validation tests the score, while direct
14 intervention tests are needed to evaluate the steering direction.

15 1 Introduction

16 Difference-of-means activation steering modifies a model’s residual stream during generation. The
17 steering direction is obtained by subtracting the mean activation of a low-property group from the
18 mean activation of a high-property group. This method has reproduced a known thermostability
19 effect in ESM2-650M [Huang et al., 2025].

20 That result leaves an important question open when the method is applied to a new protein property.
21 A strong correlation between a scoring proxy and experimental labels shows that the score contains
22 information about the property. However, it does not show that a difference-of-means vector built
23 from that property will change the model in the intended way. Four of the five properties in this
24 study reach the steering stage. The order of their proxy correlations does not match the order of their
25 steering effects.

26 We test a related assumption at the level of individual model components. An attention head
27 that correlates strongly with known protein contacts is not unusually important when it is ablated.
28 Together, these experiments distinguish evidence that a model represents a property from evidence
29 that a particular intervention controls that property.

30 **2 Related Work**

31 Huang et al. [2025] introduced difference-of-means activation steering for protein language models
32 and validated the method on thermostability. We apply the same intervention to five new properties
33 and test whether proxy accuracy predicts steering success.

34 Vig et al. [2021] identified attention heads whose attention patterns correlate with 3D protein contacts.
35 The original study described this evidence as associative. We reproduce their analysis, test the most
36 enriched head in our data by causal ablation, and extend the analysis to all 480 heads in the model.

37 Adams et al. [2025] used linear probes to identify thermostability and localization information in
38 sparse-autoencoder features from ESM-2. Their steering experiment tested whether a selected latent
39 produced more coherent sequence changes than a random latent. It did not compare the result
40 with an experimental property label. Our study addresses a related but different question: whether
41 correlational evidence predicts the result of a downstream intervention.

42 **3 Methods**

43 **3.1 Steering setup**

44 For each property, we form a steering vector from the mean activation difference between high-
45 property and low-property sequence groups. The vector is added to ESM2-650M’s residual stream
46 during generation. We evaluate the resulting sequences with a property-specific compositional proxy
47 that was checked against experimental labels before steering. Each property-derived steering direction
48 is compared with a random direction of the same norm.

49 **3.2 Evaluation protocol**

50 We fix six requirements before testing the new properties. First, the property-derived direction must
51 outperform the matched-norm random direction in a paired bootstrap test with 10,000 resamples.
52 Second, the effect must show a monotonic dose response across at least three intervention strengths
53 within a range where generation remains stable. Third, the effect must remain after the most frequently
54 substituted residues are excluded from the score. This residue-exclusion check tests whether the result
55 depends only on a narrow change in amino-acid composition. Fourth, the scoring proxy must agree
56 with experimental labels on held-out data. Fifth, the method must outperform an existing baseline
57 when a suitable baseline is available. Sixth, an evaluation of a new property must contain at least 150
58 held-out sequences.

59 We use three assessment outcomes. If proxy validation fails, we reject the proxy and do not attempt
60 steering. If the main steering test succeeds but residue exclusion fails or changes across seeds, we
61 classify the evidence as inconclusive. We call a result supported only when it satisfies every applicable
62 requirement.

63 **3.3 Attention-head ablation**

64 We also test whether contact correlation predicts the importance of an individual attention head.
65 Following Vig et al. [2021], we select the most contact-enriched head in our data, ablate it, and
66 measure the change in masked-residue prediction. We then repeat the ablation for all 480 heads in
67 the model.

68 **4 Results**

69 Table 1 summarizes the five target properties. The proxy correlations and steering effects follow
70 different orders. The strongest proxy produces the smallest measured effect, while a much weaker
71 proxy produces a larger and more reliable intervention.

72 **4.1 Binding affinity**

73 We use a ProteinGym deep-mutational-scanning assay [Notin et al., 2023] of KRAS binding to a
74 DARPIn. The proxy is built from position-specific mutational-sensitivity weights learned on the

Table 1: Results for the five target properties. Proxy r is the Pearson correlation with experimental labels on held-out data, measured before steering.

Target	Proxy r	Steering result	Assessment
Binding affinity	0.80	no measurable effect	Unsupported
Catalytic activity	0.22	repeats across 3 seeds	Supported
Intrinsic disorder	0.45	main effect 3/3; check 2/3	Inconclusive
Immunogenicity	≤ 0.10	steering not attempted	Proxy rejected
Expression yield	0.31	fails residue-exclusion check	Inconclusive

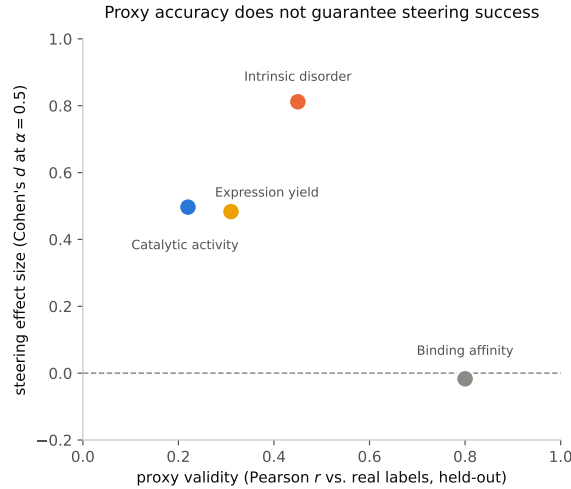


Figure 1: Proxy correlation and steering effect size for the four targets that reach the steering stage. Effect size is Cohen’s d at $\alpha = 0.5$ for the learned direction relative to the matched random direction. The four points show no positive relation between the two measurements.

75 training split. It reaches $r = 0.80$ on held-out data. The result is supported by a weight-shuffle control
76 ($r = -0.001$), evaluation on positions excluded from weight fitting ($r = 0.54$), and extrapolation
77 from single to double mutants ($r = 0.70$).

78 Despite this validation, the steering direction has no detectable effect at any tested strength. Generation
79 remains stable, so output collapse does not explain the result.

80 4.2 Catalytic activity

81 For catalytic activity, we use a cross-protein enzyme turnover dataset from DLKcat [Li et al., 2022].
82 The proxy measures the fraction of glycine minus the fraction of arginine. Its held-out correlation is
83 only $r = 0.22$, the weakest among the four targets that reach the steering stage. Even so, the steering
84 effect is monotonic, remains significant after residue exclusion, and repeats across three random
85 seeds.

86 4.3 Intrinsic disorder

87 We use DisProt annotations [Quaglia et al., 2022]. The intrinsic-disorder proxy uses the published
88 TOP-IDP amino-acid scale [Campen et al., 2008]. It reaches held-out $r = 0.45$ and per-residue AUC
89 0.71. The steering effect is significant and monotonic on all three seeds. The residue-exclusion check
90 is less stable: it passes on two seeds but fails on the third. The main effect is therefore repeatable, but
91 one artifact check is not.

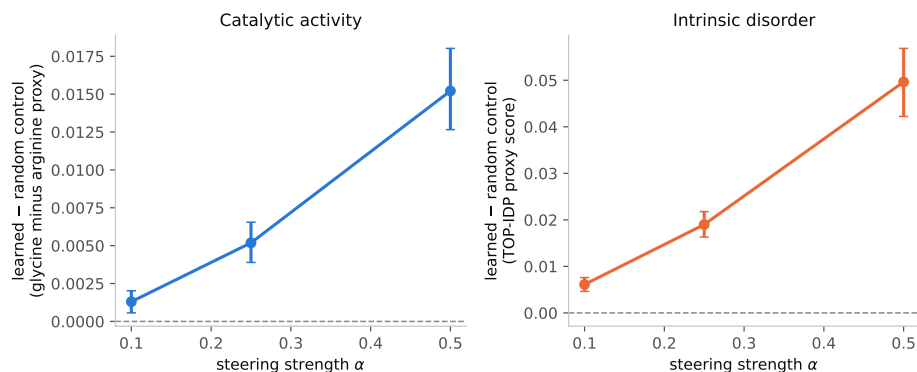


Figure 2: Dose-response curves for catalytic activity and intrinsic disorder. Both effects increase monotonically across the tested intervention strengths. Error bars show 95% bootstrap confidence intervals.

4.4 Immunogenicity and expression yield

For immunogenicity, we use records from the Immune Epitope Database [Vita et al., 2019]. Candidate proxies correlate with a surrogate binding-affinity endpoint but fall to $r \leq 0.10$ on the measured T-cell-response endpoint. One full-length-antigen proxy reaches $r = 0.38$, but organism identity explains 58% of the apparent signal. When organism identity is held out during cross-validation, the correlation changes to -0.32 . Because no proxy passes the validation requirement, we do not run a steering experiment.

We use expression-yield measurements from eSol [Niwa et al., 2009]. The proxy reaches $r = 0.31$ – 0.34 and produces a significant dose response. The effect disappears under residue exclusion, so the protocol classifies the result as inconclusive.

4.5 Attention-head ablation

The most contact-enriched head in our replication has 12.9-fold enrichment over background. Ablating it produces a change that is statistically indistinguishable from ablating a low-correlation control head. The full sweep does not provide a precise rank because the evaluation uses 104 sampled positions and many heads have identical effects. The selected head belongs to a tied block of 132 heads spanning ranks 286–417. Strong contact correlation therefore does not imply a large or unique contribution to masked-residue prediction.

5 Failure Analysis and Reliability Checks

5.1 The binding-affinity groups differ at few residues

The binding-affinity sequences are point mutants of one 188-residue reference protein. The high- and low-binding groups therefore differ by only one or two residues per sequence. Their mean per-residue compositional distance is 0.003, compared with 0.02–0.05 for the three cross-protein datasets. The proxy uses position-specific weights, so it can predict binding labels from a small number of mutations. In contrast, the steering direction averages activations over groups of nearly identical sequences. The few mutations that distinguish the groups may contribute too little to that average. This is a possible explanation for the failed intervention, not a demonstrated mechanism.

5.2 The expression-yield direction overlaps with disorder

We compare the expression-yield steering vector with every other target vector. Its cosine similarity with the intrinsic-disorder direction is $+0.38$ across all 33 layers and rises to $+0.56$ – 0.67 in the three deepest layers. This overlap shows that the directions are not orthogonal and may capture some of the same compositional variation. It does not show that disorder causes the expression-yield

123 result. Because the expression-yield result also fails residue exclusion, we do not count it as separate
124 evidence for a new steering effect.

125 **5.3 Checks that changed the interpretation**

126 An early thermostability comparison selected the best intervention strength from a range that included
127 unstable generation. At the selected strength, the 33-layer intervention had already collapsed, while a
128 5-layer intervention had not. The 5-layer version appeared better only because it failed at a higher
129 strength. After we restricted the comparison to a stable range and reran the experiment, the 5-layer
130 intervention retained 43–59% of the full effect.

131 The intrinsic-disorder experiment revealed a different problem. The main effect repeats across all
132 three seeds, but the residue-exclusion check passes on only two. A single run could therefore give a
133 misleading verdict about that criterion.

134 **6 Discussion**

135 These results show why representation and intervention should be evaluated separately. A proxy can
136 accurately summarize experimental labels without defining a direction that changes model output in
137 the intended way. Likewise, an attention head can align with a biological pattern without making a
138 large, individually measurable contribution to the model’s prediction.

139 Proxy validation remains necessary because it rules out scores that do not track the target, as the
140 immunogenicity experiment shows. The problem arises when evidence about the score is also treated
141 as evidence about the steering vector. The binding-affinity experiment shows that the score and the
142 vector can depend on different sources of signal.

143 The experiments suggest three practical checks. Before constructing a steering vector, researchers
144 should measure how much the high- and low-property groups differ. Comparisons between interven-
145 tions should use the same range of strengths, and every intervention should produce stable sequences
146 throughout that range. Important results should be repeated across seeds, including the artifact checks
147 used to support them.

148 **7 Limitations**

149 All experiments use ESM2-650M at one model scale. The proxies are compositional heuristics
150 rather than learned predictors. A stronger non-compositional proxy could behave differently. We
151 selected each final proxy from several candidates using the same held-out split on which we report its
152 correlation, so the reported correlations are likely optimistic.

153 We ran multiple significance tests without a formal correction. Requiring agreement across interven-
154 tion strengths and seeds reduces this risk but does not remove it. Seed robustness was checked for
155 only two targets. We also did not evaluate generated sequences with a wet-lab assay or a structural
156 plausibility model. A successful protocol result means that the intervention changes a validated proxy.
157 It does not establish that the generated protein has the intended biological property.

158 **8 Conclusion**

159 Across these experiments, higher proxy accuracy does not guarantee more successful steering. The
160 binding-affinity result provides the clearest counterexample, and the attention-head ablation shows a
161 related mismatch at the level of a model component. Correlational evidence can show that a model
162 represents a property, but direct intervention tests are still needed before that representation is used to
163 guide steering.

Responsible-use statement

This work studies errors in the evaluation of activation steering for protein language models. It does not introduce a new method for generating harmful biological sequences. The target properties are standard protein-engineering objectives, and the experiments use a previously published steering method. We do not identify risks beyond those already associated with protein language models and standard protein-engineering research.

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